

Instagram: Improving Engagement Quality & Retention

1. INTRODUCTION

Instagram is one of the most popular social media platforms, where people connect, share, and explore content through features like Feed, Reels, Stories, and Messages. For a long time, success on such platforms was measured by how much time users spent on the app. However, more time does not always mean a better or healthier experience.

Today, the focus is shifting toward engagement quality — how meaningfully users interact with content rather than just scrolling. Actions like commenting, sharing, and messaging reflect deeper engagement compared to passive viewing. Improving engagement quality is important for user well-being, long-term retention, and overall platform growth.

This project analyzes user behavior and lifestyle data to understand what factors are linked to higher engagement quality on Instagram. The goal is to identify patterns that encourage more meaningful interactions and support healthier platform usage.

2. PROBLEM STATEMENT

Instagram serves millions of users daily across features like Feed, Reels, Stories, and Messages. While overall activity on the platform is high, spending more time does not always mean users are engaging in a meaningful or healthy way. The focus is now shifting from maximizing time spent to improving the quality of engagement. The main challenge is to understand which user behaviors lead to high-quality engagement and which are linked to passive, low-intent usage. Factors such as session patterns, interaction types, and lifestyle signals may influence engagement quality. Identifying these patterns can help improve user experience, reduce unhealthy usage, and support long-term retention.

3. ANALYSIS APPROACH

This project follows a structured approach to understand and improve engagement quality on Instagram.

1. Understanding the Business Problem – The focus is on improving the quality of engagement rather than just increasing time spent on the platform.
2. Defining Metrics and Hypotheses – Engagement Quality Efficiency (EQE) is used as the main metric, along with supporting metrics, to test how user behavior and lifestyle factors affect engagement.

3. Data Cleaning and Preparation – The data was checked for quality, outliers were treated, and new features such as EQE and engagement segments were created.
4. Exploratory Data Analysis (EDA) – Patterns in EQE were analyzed across demographics, usage behavior, content preferences, and lifestyle factors using visualizations.
5. User Segmentation and Supporting Analysis – Users were grouped into Low, Medium, and High EQE categories to compare behaviors, along with interaction health and ad efficiency metrics.
6. Insights and Recommendations – Findings were translated into actionable recommendations to promote healthier and more meaningful engagement.

4. BUSINESS CONTEXT & OBJECTIVE

Business Context

Instagram is one of the world's largest social platforms, used daily for entertainment, discovery, and communication across formats like Feed, Reels, Stories, Explore, and DMs. As the platform grows, the focus is shifting from maximizing time spent to improving engagement quality, ensuring user activity is meaningful and healthy rather than driven by passive scrolling or notification overload. Poor engagement quality can reduce long-term retention, weaken the creator ecosystem, increase trust risks, and lower monetization effectiveness. To address this, Instagram relies on user behavior and lifestyle data to better understand how meaningful engagement happens on the platform.

Objective

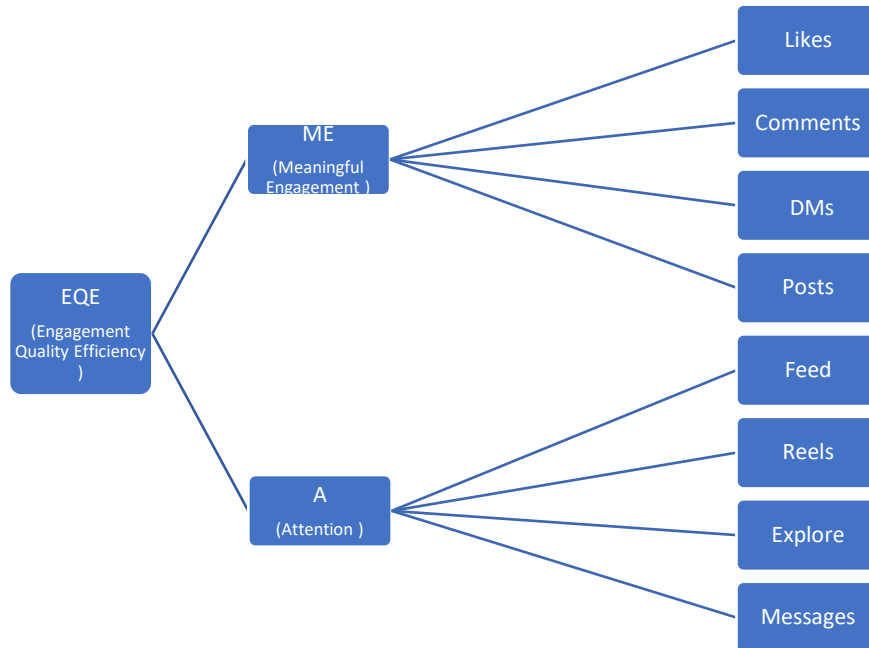
The objective of this project is to analyze user behavior and lifestyle data to identify the factors that influence Engagement Quality Efficiency (EQE). The goal is to distinguish between meaningful engagement and passive usage, recognize patterns linked to high and low engagement quality, and detect early signs of dormancy. These insights aim to support healthier user experiences and sustainable platform growth.

5. DATASET

This dataset contains over 1.55 million user records and 58 features, covering interaction activity (likes, comments, DMs, posts), time spent across different sections of the app, session patterns, content preferences, notification behavior, account settings, and lifestyle factors such as sleep, exercise, and stress levels. Each row represents a user's activity over a defined time period, allowing analysis of how behavioral and lifestyle patterns relate to engagement quality.

6. Metrics and Hypothesis

- Metric Tree



To evaluate engagement quality on Instagram, this project defines Engagement Quality Efficiency (EQE) as the primary metric. **$EQE = \text{Meaningful Engagement} \div \text{Attention (Time Spent)}$**

Meaningful Engagement (ME) includes active interactions such as likes, comments, direct messages, and posts. Attention (A) represents the total time spent across different areas of the platform, including Feed, Reels, Explore, and Messages.

In addition to EQE, several supporting metrics were used:

Interaction Health Metrics

- Comment-to-Like Ratio – Measures the depth of interaction beyond simple likes
- DM Activity Index – Measures average daily messaging activity

Ad Efficiency Metrics

- Ad CTR Proxy – Ratio of ads clicked to ads viewed
- Ad Load Proxy – Number of ads viewed relative to total attention

- Hypothesis

H1: Older users have higher engagement quality

H2: More sessions per day leads to lower engagement efficiency (doom-scrolling pattern)

H3: Shorter sessions are more meaningful than long sessions

H4: Engagement quality differs significantly by gender

H5: Engagement quality differs by content type, privacy setting, or subscription

H6: Healthier lifestyle habits lead to better engagement quality

H7: High EQE users are more valuable for ads

H8: A small segment of users shows dormancy risk

7. DATA CLEANING AND PREPARATION

1. Initial Data Assessment

The dataset contains 1,547,896 rows and 58 columns.

- Duplicate rows: 0
- Missing values: 0 across all columns

2. Data Type Correction

The column `last_login_date` was originally stored as an object and was converted to datetime format to enable time-based analysis.

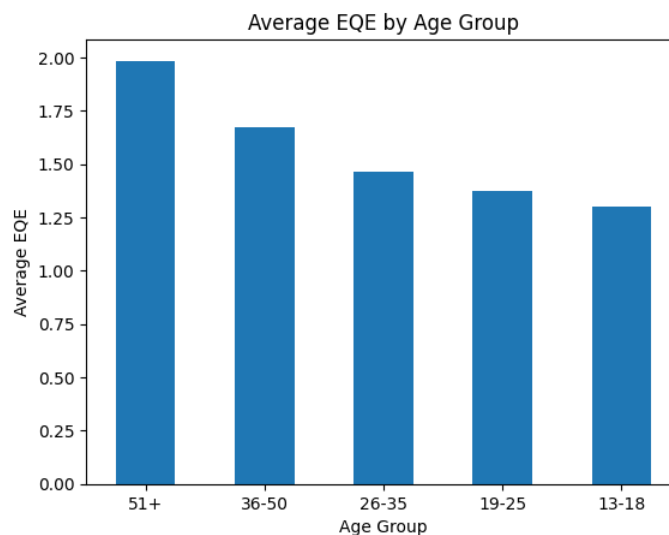
3. Outlier Detection and Treatment

Outliers were treated using the IQR method through value capping. Since the dataset size is large and values were clipped rather than removed, only a small fraction of extreme values were adjusted, preserving overall data integrity.

8. EXPLORATORY DATA ANALYSIS (EDA)

1. Demographic Analysis

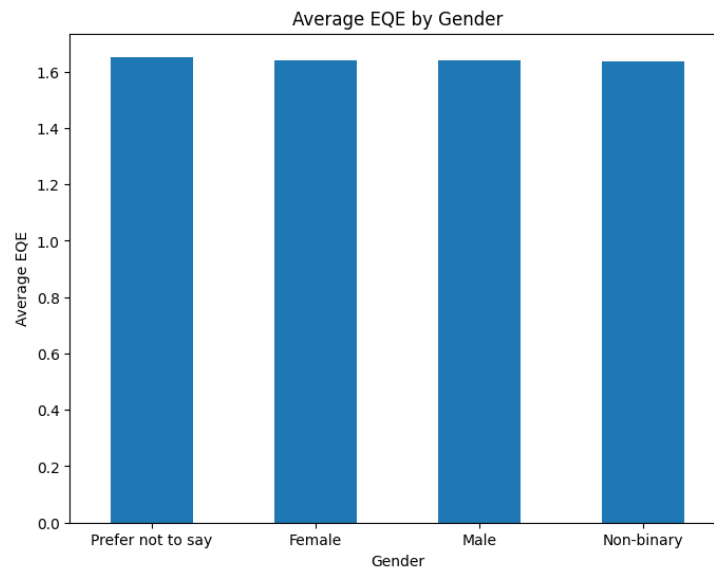
a. EQE by age



A bar chart was used to compare EQE across age groups, as it clearly shows differences between categories. Older users (51+) have the highest EQE, while younger users (13–18) show lower engagement efficiency. This suggests that younger users may be engaging more passively.

Instagram can introduce features that encourage interactive behaviors like commenting or sharing among younger users.

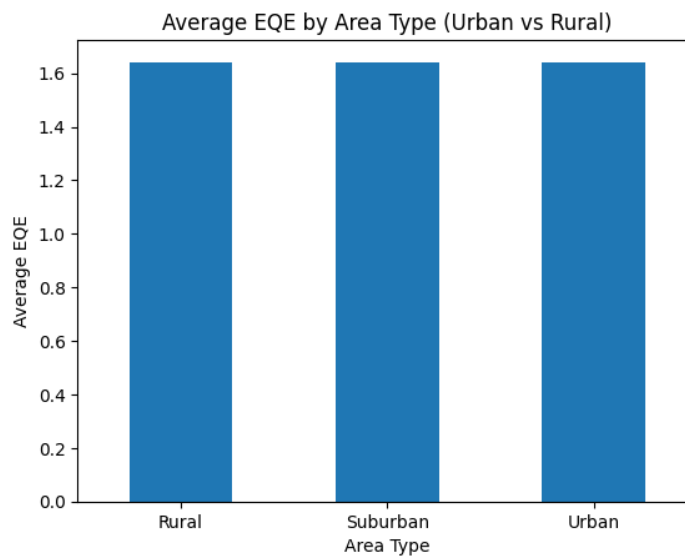
b. EQE by gender



The bar chart shows that EQE is nearly the same across all gender groups. There are only minor differences, indicating that engagement quality does not significantly vary by gender.

Insight: Gender does not strongly influence engagement quality; user behavior plays a bigger role.

c. EQE by Urban vs Rural

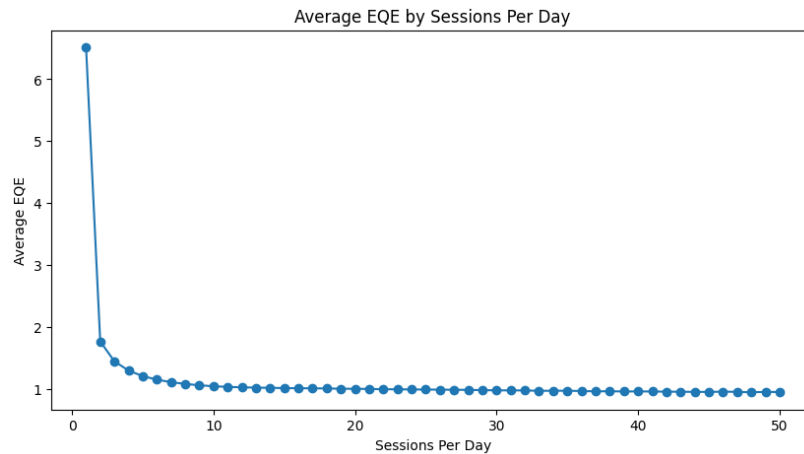


The chart shows that EQE values are nearly the same for rural, suburban, and urban users. The differences are very small, indicating that where a user lives does not significantly affect engagement quality.

Insight: Engagement behavior depends more on how users interact with the platform than where they live.

2. Platform Behavior

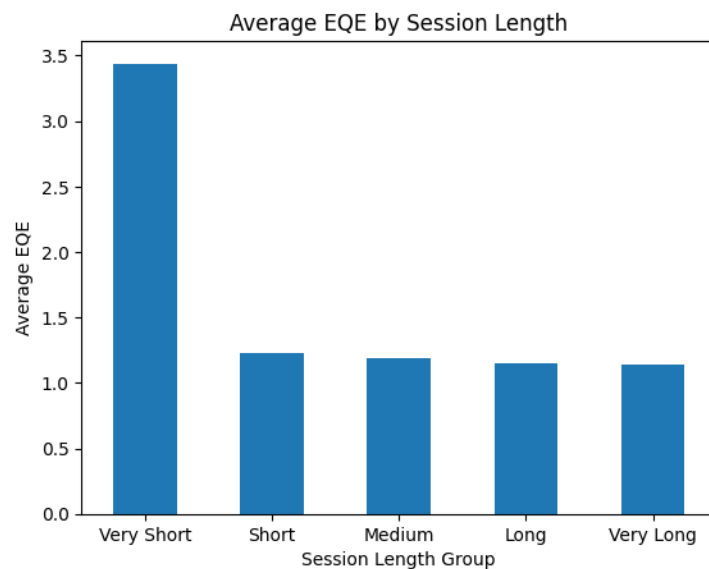
a. Sessions per day



A line chart was used to examine how Engagement Quality Efficiency (EQE) varies with the number of daily sessions. The results show that EQE is highest for users with only one session per day (6.51) and drops sharply as session frequency increases. From two sessions onward, EQE stabilizes at a much lower level.

Insight: Frequent app visits are linked to lower engagement efficiency, suggesting more passive or habitual usage.

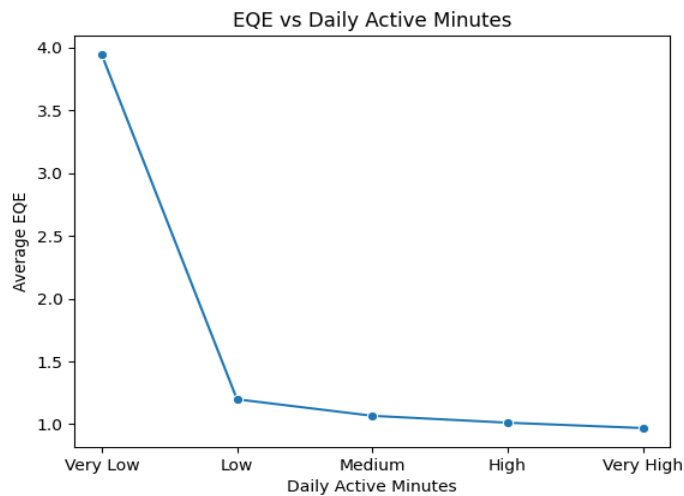
b. Average session length



Users were grouped based on session duration. The highest EQE is observed among users with very short sessions (3.44), while EQE gradually declines as session length increases, reaching the lowest levels for long and very long sessions.

Insight: Short sessions are more focused and meaningful, while longer sessions tend to reflect passive browsing.

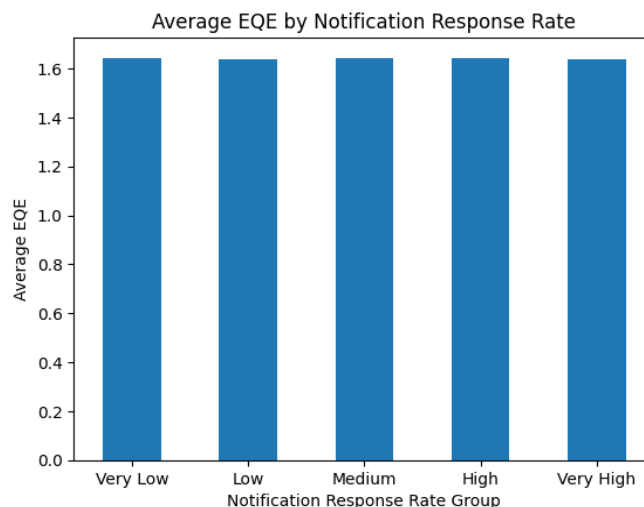
c. Daily active minutes on Instagram



A line chart shows how Engagement Quality Efficiency (EQE) changes with daily active minutes. EQE drops sharply from very low to low usage and then gradually declines as time spent increases.

Insight: Users who spend less time on Instagram engage more meaningfully per minute. As usage increases, engagement becomes less efficient, suggesting a shift toward more passive browsing rather than active interaction.

d. Notification response rate

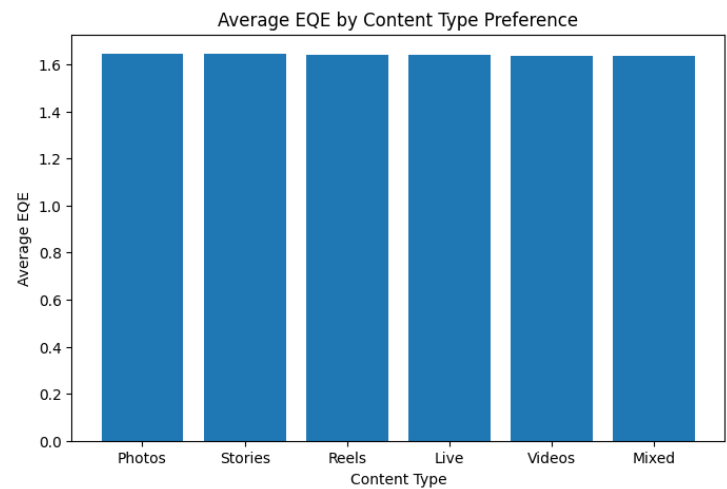


EQE remains almost the same across all notification response rate groups, from Very Low to Very High.

Insight: Responding to notifications more often does not significantly change engagement quality. This suggests that meaningful engagement is driven more by how users interact inside the app rather than how frequently they react to notifications.

3. Content Preferences

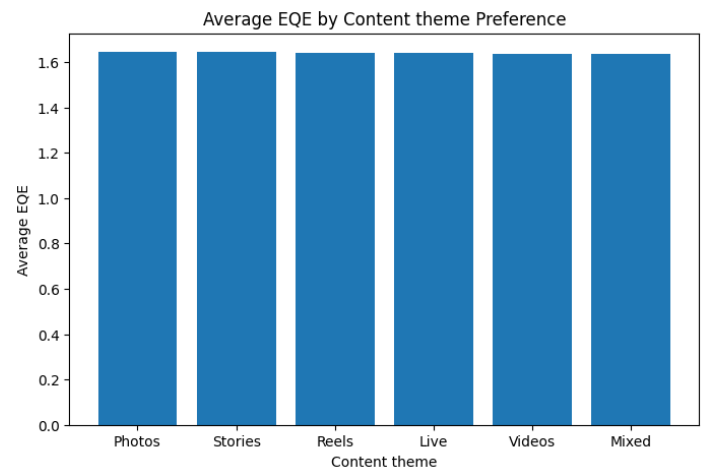
a. Content type preference



A bar chart was used to examine how Engagement Quality Efficiency (EQE) varies across different content type preferences. The results show that EQE remains almost identical for users who prefer Photos, Stories, Reels, Live videos, Videos, or Mixed content formats, with only very minor differences in values.

Insight: Engagement quality does not significantly depend on the type of content users prefer, suggesting that user behavior patterns matter more than content format.

b. Preferred content theme

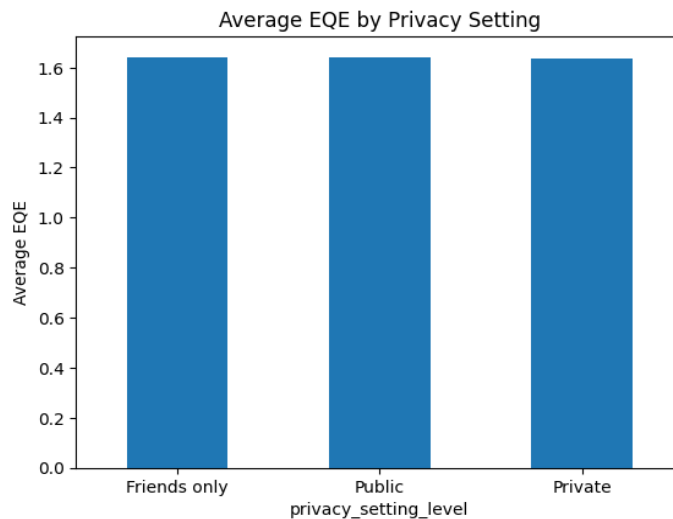


A bar chart was also used to analyze how EQE varies across different content theme preferences. The results indicate that EQE values are very similar across all content themes, with no theme showing a noticeably higher or lower engagement efficiency.

Insight: Content themes have minimal impact on engagement quality, reinforcing that meaningful engagement is driven more by how users interact with the platform rather than what topics they consume.

4. Account & Trust Factors

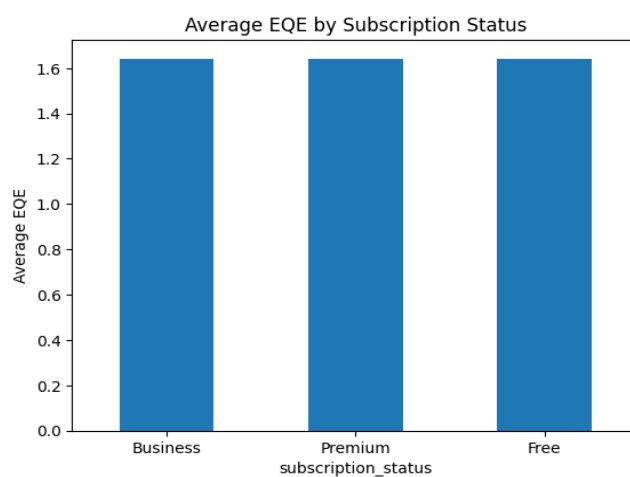
a. Privacy setting level



A bar chart was used to compare Engagement Quality Efficiency (EQE) across different privacy setting levels. The results show that EQE values are almost identical for Public, Private, and Friends-only accounts, with no meaningful variation between groups.

Insight: Privacy settings do not appear to influence engagement quality. Users engage with similar efficiency regardless of how open or restricted their profiles are.

b. Subscription status

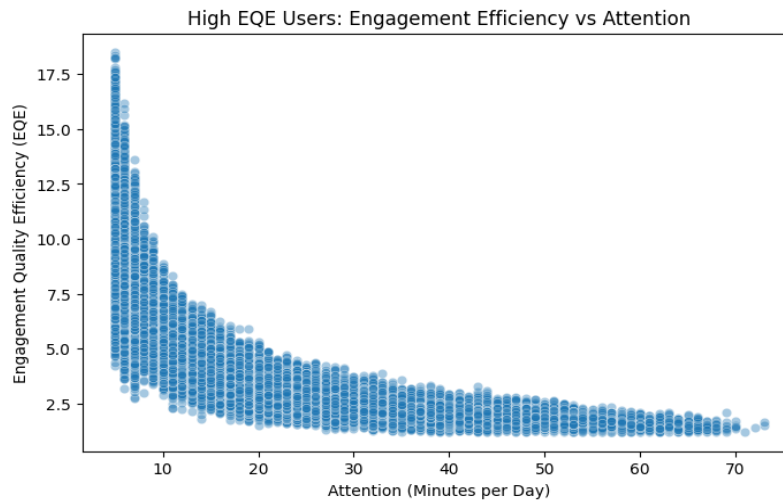


A bar chart was used to analyze how EQE varies across subscription types (Free, Premium, Business). The chart shows nearly identical EQE values for all subscription categories, with no significant difference in engagement efficiency.

Insight: Paying for premium or business features does not necessarily translate into more meaningful engagement. Engagement quality seems driven more by user behavior than subscription tier.

5. Identify Which Combinations Drive Behavior

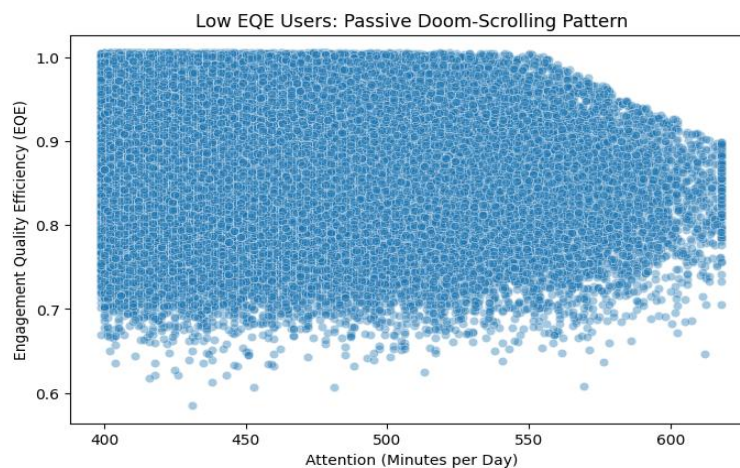
a. High EQE (healthy engagement)



A scatter plot was used to analyze the relationship between Engagement Quality Efficiency (EQE) and attention (time spent) among high EQE users. The plot shows that EQE values are highest at lower attention levels and gradually decline as time spent increases.

Insight: High engagement quality is achieved during short, focused sessions. Increasing time spent does not improve engagement efficiency, indicating that meaningful engagement is driven by intentional usage rather than prolonged browsing.

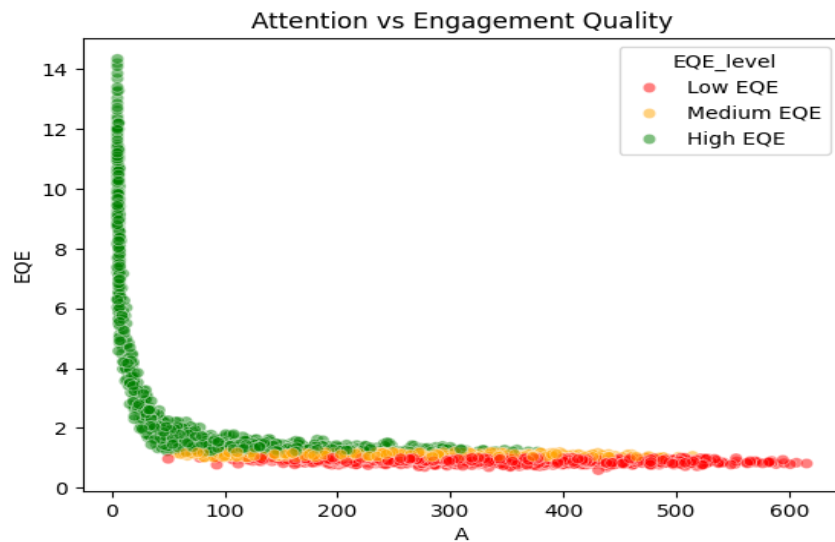
b. Low EQE (passive doom-scrolling)



A scatter plot shows how EQE changes with time spent among Low EQE users. Despite high attention levels, engagement efficiency stays low.

Insight: These users spend a lot of time on Instagram but interact less meaningfully, suggesting passive scrolling rather than intentional engagement.

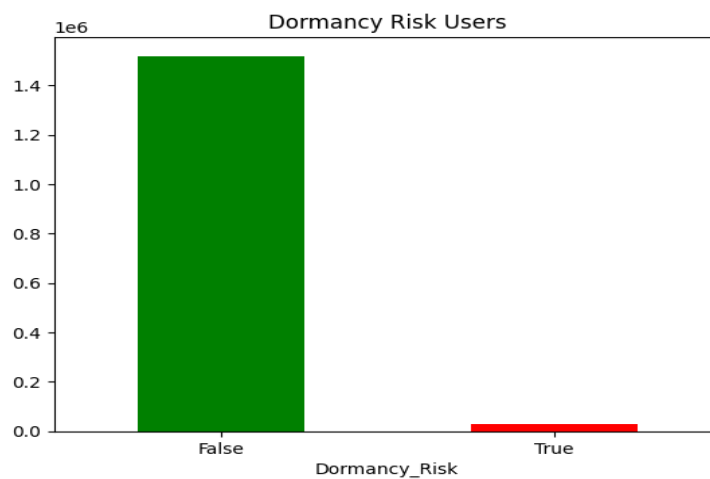
c. High EQE vs Low EQE



This scatter plot compares time spent and engagement quality for different user groups. High EQE users spend less time but interact more meaningfully, while Low EQE users spend more time with lower engagement efficiency.

Insight: More time spent on the app does not guarantee better engagement. Meaningful engagement is linked to focused usage rather than prolonged browsing.

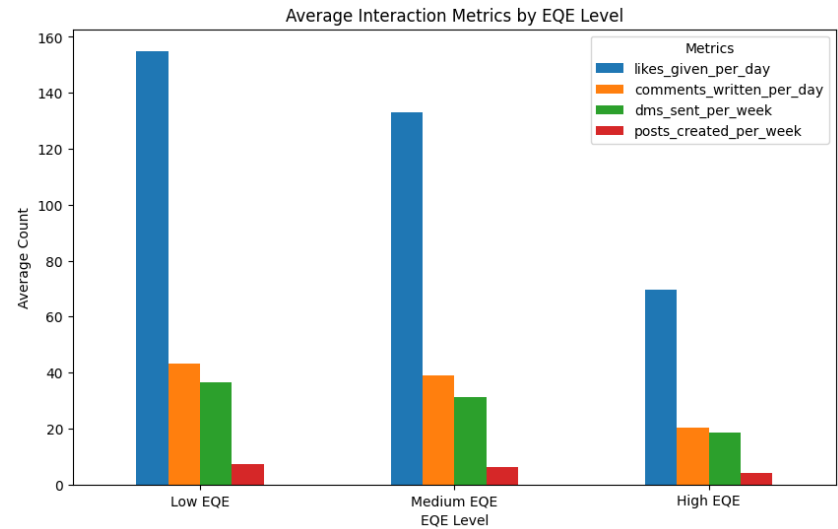
d. Dormancy risk users



This bar chart shows how many users are at risk of becoming inactive. Most users are active, but a small group shows signs of low engagement and possible future dropout.

Insight:Even though the at-risk group is small, they may stop using the app if not supported, so early re-engagement is important.

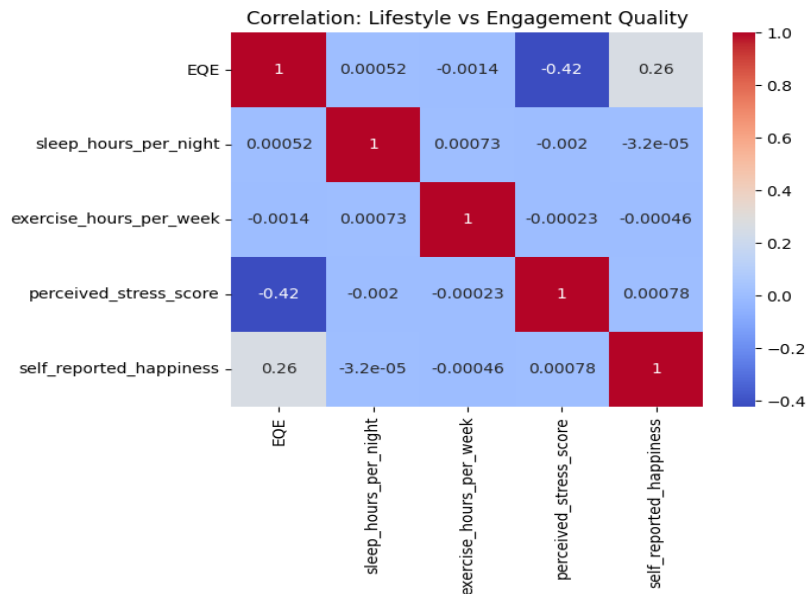
e. User Interaction Behavior by EQE Level



This chart shows how user actions differ across engagement quality groups. Low EQE users do more total actions, but their engagement is less meaningful because they spend much more time on the app.

Insight:High EQE users may do fewer actions, but their interactions are more meaningful and efficient, showing better engagement quality.

6. Lifestyle & Engagement Quality

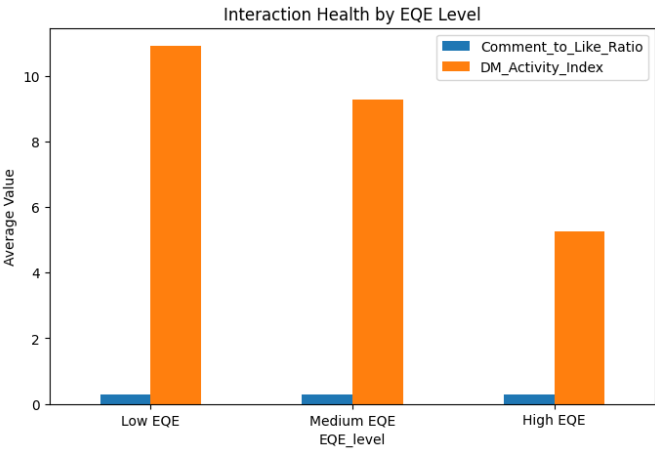


This heatmap shows how lifestyle factors relate to engagement quality. Stress is linked to lower EQE, while happiness is linked to higher EQE, and sleep and exercise have little effect.

Insight:Users’ emotional well-being appears to have a stronger connection to engagement quality than physical habits, suggesting mood may influence how meaningfully users interact on Instagram.

7. Supporting Metrics Analysis

a. Interaction Health Analysis

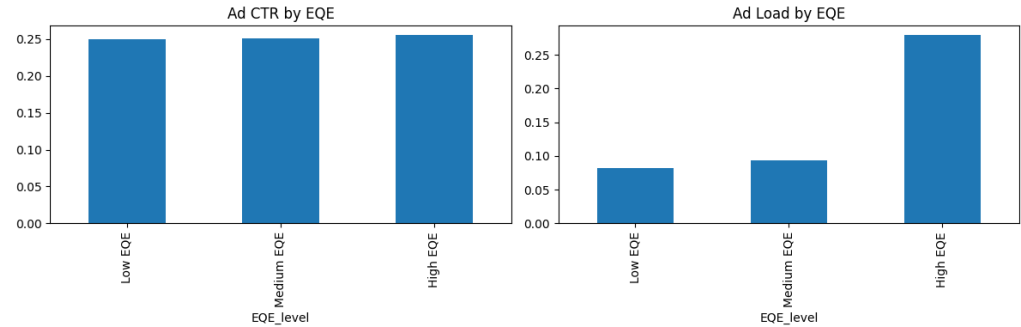


This chart compares how users interact at different engagement quality levels. High EQE users comment more compared to likes, showing deeper interaction, while Low EQE users send more messages but with lower engagement quality.

Insight:

High EQE users focus on meaningful interactions, while Low EQE users are more active but less efficient in how they engage.

b. Ads Efficiency Analysis



These charts show how users respond to ads. High EQE users click on ads slightly more, but they also see more ads than other users.

Insight:

Users who engage more meaningfully are also more responsive to ads, but showing them too many ads could affect their experience, so balance is important.

9. Hypothesis Testing Results

Hypothesis	Result	Conclusion
H1: Older users have higher engagement quality	Supported	Older users show higher EQE, indicating more meaningful interaction patterns.
H2: More sessions per day leads to lower engagement efficiency (doom-scrolling pattern)	Supported	High session frequency is linked to lower EQE, suggesting frequent but shallow engagement.
H3: Shorter sessions are more meaningful than long sessions	Supported	Very long sessions are associated with lower EQE, while shorter sessions show higher engagement efficiency.
H4: Engagement quality differs significantly by gender	Not Supported	EQE remains nearly consistent across gender groups.
H5: Engagement quality differs by content type, privacy setting, or subscription	Not Supported	These factors do not show strong influence on engagement efficiency.
H6: Healthier lifestyle habits lead to better engagement quality	Partially Supported	Lower stress levels are linked to higher EQE, while sleep and exercise show weaker impact.
H7: High EQE users are more valuable for ads	Weakly Supported	High EQE users show slightly better ad responsiveness but differences are small.
H8: A small segment of users shows dormancy risk	Supported	Around 2% of users show inactivity signals linked to low engagement.

10. KEY INSIGHTS & RECOMMENDATIONS

Key Insights

- Engagement quality declines with increased usage intensity.**
Users with more sessions per day, longer session lengths, and higher daily active minutes consistently show lower EQE, indicating passive or habitual usage rather than meaningful engagement.
- High-quality engagement happens in short, focused sessions.**
High EQE users achieve meaningful interactions while spending less time on the platform, showing that intentional usage leads to better engagement efficiency.
- Demographics and preferences have limited impact on EQE.**
Gender, location (urban/rural), content type, content theme, privacy settings, and subscription status show minimal differences in engagement quality, suggesting behavior matters more than user attributes.

4. **Notifications do not improve engagement quality on their own.**
EQE remains almost unchanged across notification response rate groups, indicating that bringing users back via notifications does not guarantee meaningful interaction.
5. **Low EQE users show passive doom-scrolling behavior.**
These users spend a large amount of time on Instagram but maintain low engagement efficiency, confirming that more attention does not translate into better engagement.
6. **Lifestyle factors show a limited but important relationship.**
Higher stress levels are moderately associated with lower EQE, while happiness is positively associated with engagement quality. Sleep and exercise show weak relationships.
7. **A small segment of users shows dormancy risk.**
A minority of users display low session frequency, low notification response, and long gaps since last login, indicating early signs of disengagement.
8. **High EQE users are slightly more valuable for ads.**
High EQE users show marginally better ad responsiveness, but they are also exposed to a higher ad load.
9. **Engagement quality increases slightly with age.**
Older users (especially 51+) show higher average EQE compared to younger users, suggesting they engage more intentionally, while younger groups tend to have more passive usage patterns.

Recommendations

1. **Shift success metrics from time spent to engagement quality.**
Product decisions should prioritize meaningful actions (comments, shares, posts, DMs) over increasing total usage time.
2. **Encourage short, intentional sessions.**
Design features and UI nudges that promote focused interactions rather than prolonged scrolling.
3. **Rethink notification strategy.**
Notifications should be personalized based on past meaningful engagement, not just activity frequency.
4. **Identify and manage passive usage early.**
Users with high time spent and low EQE should be monitored as a risk group and guided toward more interactive behaviors.
5. **Support user well-being.**
Given the link between stress and lower engagement quality, light well-being prompts or content balance mechanisms may help promote healthier usage.
6. **Target dormancy-risk users with early interventions.**
Users showing inactivity signals should receive timely re-engagement efforts to reduce future churn.
7. **Balance ad exposure for high EQE users.**
Since high EQE users respond better to ads but see more of them, ad load should be carefully managed to maintain engagement quality.

11.CONCLUSION

This analysis shows that spending more time on Instagram does not guarantee meaningful engagement. High engagement quality is linked to shorter, more intentional sessions, while heavy usage often reflects passive browsing. Focusing on interaction quality rather than time spent can help improve user well-being, retention, and long-term platform growth.